

Unit 3, Lesson 8: Solving Multi-Step, Real-World Problems with Fractions or Decimals – Part 1

Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

Learning Target

- ☐ I can solve three-step, real-world problems with decimals or fractions.

3.8.1 Warm-Up: Career – Body Structures Engineer

Rebecca Lees, a body structures engineer from Jaguar Land Rover, works on high-profile projects and talks about how she came to pursue a career in automotive, her educational background, and how an international work placement played a key role in setting her on a path to her career.



1. What did you find interesting about body structures engineers?

2. Structural engineers focus on the structure of a building or a bridge. The structures should withstand the stresses and pressures of the environment. What stresses or pressures should the body of a car be able to withstand?

3.8.2 Guided Instruction: Three-Step Rational Problems

1. For his recital, a student practiced the piano 2.18 hours each day for 5 days last week. This week, he practiced 0.75 hours each day for 4 days. How many hours did he spend practicing?
2. The Across Florida 200 Trail Run is an event where participants run 200 miles of trails between the Withlacoochee Bay Trail and Crescent Beach. To prepare for the Across Florida 200 Trail Run, Walt ran 2.18 hours each day for 5 days last week. This week, he ran 2.75 hours each day for 4 days. Walt runs at an average pace of 6.23 miles per hour.

a. How many hours has Walt run over the last two weeks?

b. How many miles has Walt run over the last two weeks?

c. Based on Walt’s average pace, how much longer would it take him to run the Across Florida 200 Trail Run than the amount of time he has spent practicing the last two weeks?
3. The Across Florida 200 Trail Run offers a team competition where a team of 4 runners take turns to run the 200 miles.

A team of runners and their average pace is shown.

Name	Molly	Reginald	Tyler	Liz
Average Pace (miles per hour)	6.75	6.75	8.2	9.58

To prepare for the race the team ran part of the trail.

Team Member	Description	Miles
Tyler	Withlacoochee Bay Trail to Lake Rousseau Dam & Recreational Area	10
Liz	Lake Rousseau to Florida Trail	15
Molly	Florida Trail Part 1	35
Reginald	Florida Trail Part 2	25

How long did it take the team to run the practice run of 85 miles?

3.8.3 Your Turn!

1. On Sunday, a farmer picked $1\frac{3}{4}$ barrels of blueberries. On Monday, the farmer picked $\frac{1}{5}$ as many blueberries as he did on Sunday. On Tuesday, the farmer discovered that half of his barrels of blueberries were spoiled. How many total barrels of blueberries did the farmer have left to sell?
2. A pool that holds 273.68 gallons of water lost 4.79 gallons of water each day while there was a leak that went unnoticed. Tad noticed the water was lower, so he added 20.45 gallons of water before he realized there was a leak on the side. If Tad fixed the leak after 8 days of leaking, how many gallons of water does he currently have in the pool?
3. A paving company ordered $4\frac{3}{4}$ tons of cement to pave the streets of a new neighborhood. Additionally, the company ordered $7\frac{2}{3}$ tons of cement to pave the guest parking lot for the pool area at the front of the neighborhood. After pouring, they only needed $\frac{2}{3}$ of the cement they ordered. How many tons of cement were left unused?
4. A grocery store opens a new wing bar that charges by the pound. Yasmine wants to purchase $\frac{1}{4}$ pound of garlic parmesan wings, $\frac{1}{8}$ pound of mild wings, and $\frac{2}{5}$ pound of teriyaki wings. She also wants to purchase some hot wings. If Yasmine wants to purchase a total of 2 pounds of wings, how many pounds of hot wings should she buy?
5. Rawslyn is creating study kits for her students. Each folder costs \$1.79 and each copy of the study guide costs \$0.95. Each student gets one of each and there are 22 students in the class. If Rawslyn has \$35.43 in her wallet, how much more money will Rawslyn need to create the study kits?

3.8.4 Wrap-Up: Cool Down

1. For each statement, write whether you agree with it, disagree with it, or are unsure.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	
When you add two positive decimals together, the value is always larger.	
To add fractions, their denominators must be the same.	
The product of two fractions is always a larger number.	
When you put a zero on the end of any number, it makes it larger.	